Ex.No.5

Program :

```
#include <stdio.h>

struct priority_scheduling
{
   char process_name;
   int burst_time;
   int waiting_time;
   int turn_around_time;
   int priority;
};

int main()
{
   int number_of_process, total = 0;
   struct priority_scheduling temp_process;
   int ASCII_number = 65;
   int position;
   float average_waiting_time, average_turnaround_time;

   printf("Enter the total number of Processes: ");
   scanf("%d", &number_of_process);

   struct priority_scheduling process[number_of_process];

   printf("\nPlease Enter the Burst Time and Priority of each process:\n");

   for (int i = 0; i < number_of_process; i++)
   {
      process[i].process_name = (char)ASCII_number;
```

```
    printf("\nEnter the details of the process %c\n", process[i].process_name);

    printf("Enter the burst time: ");
    scanf("%d", &process[i].burst_time);

    printf("Enter the priority: ");
    scanf("%d", &process[i].priority);

    ASCII_number++;
}

for (int i = 0; i < number_of_process; i++)
{
    position = i;

    for (int j = i + 1; j < number_of_process; j++)
    {
        if (process[j].priority > process[position].priority)
            position = j;
    }

    temp_process = process[i];
    process[i] = process[position];
    process[position] = temp_process;
}

process[0].waiting_time = 0;

for (int i = 1; i < number_of_process; i++)
{
```

```
    process[i].waiting_time = 0;

    for (int j = 0; j < i; j++)
    {
        process[i].waiting_time += process[j].burst_time;
    }

    total += process[i].waiting_time;
}

average_waiting_time = (float)total / number_of_process;
total = 0;

printf("\n\nProcess Name\tBurst Time\tWaiting Time\tTurnaround Time\n");

for (int i = 0; i < number_of_process; i++)
{
    process[i].turn_around_time = process[i].burst_time + process[i].waiting_time;
    total += process[i].turn_around_time;

    printf("%c\t\t%d\t\t%d\t\t%d\n",
            process[i].process_name,
            process[i].burst_time,
            process[i].waiting_time,
            process[i].turn_around_time);
}

average_turnaround_time = (float)total / number_of_process;

printf("\nAverage Waiting Time : %f", average_waiting_time);
printf("\nAverage Turnaround Time : %f\n", average_turnaround_time);
```

```
    return 0;
}
```

Output Clear

```
Enter number of process:3
nEnter Burst Time:
p1:3
p2:2
p3:6
nProcesst Burst Time tWaiting TimetTurnaround Time
np2tt 2tt 0ttt2np1tt 3tt 2ttt5np3tt 6tt 5ttt11nnAverage Waiting Time=2
    .333333nAverage Turnaround Time=6.000000n

=== Code Execution Successful ===
```